

A Review of Mechanisms and Models Accounting for Surface Potential Decay

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Abstract—We present here a wide review of the existing models accounting for surface potential decay, developed in various areas of electrostatics. Several polarization or transport processes may be involved. Though most of the models initially stemmed from electrostatics and semiconductor physics, around the notion of mobility, experiments on polymers often require “thermodynamic” models, describing progressive charge detrapping. Different physical processes likely to be involved in the potential decay (dipolar relaxation, induced conductivity, dispersive transport, slow detrapping) can lead in disordered materials to the same time response, the challenge being to design inventive procedures to distinguish them. A particular attention will be given to the application of these models to polyimide and fluorinated ethylene propylene. The specific case of radiation-induced conductivity in these materials will also be examined in detail.

Index Terms—Charge injection, dielectric polarization, internal charging, modeling, nondestructive testing, potential decay.

I. BRIEF OVERVIEW

SURFACE and internal charging of dielectric materials is a potential cause of surface discharges and satellite anomalies, due to the fluctuating irradiation levels induced by space environment [1]. High charging potentials have been measured in orbit on various materials, as in the SCATHA satellite experiments [2].

Understanding the conduction mechanisms and reducing the charging levels are therefore important industrial issues for satellite designers and manufacturers. Surface potential measurements under irradiation and after charging (potential decay) are the most significant laboratory tests to qualify and understand the charging and discharging behavior of insulating materials [3]–[5].

We will focus here on the mechanisms of charge dissipation occurring in an insulator after charging. The space environment has its specific features, very different from earth’s usual laboratory or industrial conditions. However, many of the possible dissipation mechanisms involved in the materials used in space have been described in other contexts, and the interpretation of the surface potential decay data requires a wide knowledge of these mechanisms. The purpose of this paper is therefore to

provide a rapid overview of surface potential decay studies, relying on the analysis of a bibliography of more than 250 papers. The present work cannot, however, pretend to exhaustiveness, as the electrostatic probe worldwide diffusion since the 1960s and 1970s has led to numerous works throughout the world. We limited our study to papers presenting sizeable modeling, or experimental results which notably influenced model development.

Four main fields of industry led to most of the studies involved in our review, with their specific concerns. Electrical engineering, as space industry, aims at reducing as much as possible the damages due to inner space charge. Concerning electrets, the challenge is, on the opposite, to stabilize a static charge during a very long time, whereas electrooptics developed devices where the charge level and dissipation have to be thoroughly controlled.

A. Electrical Engineering

The multiform development of insulating polymers for high voltage insulation has been the source of numerous works on surface potential in Japan [6]–[8], France [9]–[16] or Canada [17]–[21]. An experimental result was issued from these studies which had an important influence on potential decay models: the observation published by Ieda in 1967 [6], that, on polyethylene, in particular circumstances, potential decay curves obtained on the same material with several initial charge levels did cross each other (Fig. 1). This cross-over phenomenon has been the scope of many works and excited theoreticians’ imagination [9], [17], [19], [22], [23]. The most convincing explanation of the phenomenon has in the end been published by Perlman et al. from Quebec [19].

B. Electrets

The feasibility to store charges for a long time in insulating materials has been exploited to develop many electret-based devices, in the field of electroacoustics, sensors, filtration, etc. Charge decay mechanisms were studied here from an opposite point of view, which was to improve the time stability of the charge. Therefore, these teams, among the researchers from Darmstadt [24]–[26] and Sao Paulo [22], [27]–[29], concentrated on the topic of charge build-up and trapping into the material, rather than on decay models, even if they published a few of them (see for instance [24], [29] for fluorinated ethylene propylene (FEP), or [25] for SiO₂).

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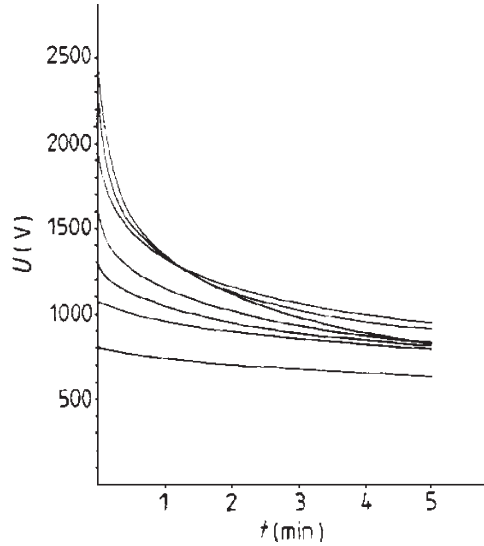


Fig. 1. Potential decay curves crossover [23].

C. Electrooptics

The design of photocopiers and laser printers led to much of groundwork, issued by American industrial research laboratories, as IBM [30], Xerox [31], [32], or Kodak [33], on corona charging, charge injection mechanisms, and potential decay dynamics, particularly on photoconductive insulators. Taking semiconductor physics as a starting point, the decrease of the potential due to transport mechanisms in a disordered material has been modeled in detail, by introducing trapping, which led to the dispersive transport models and to the thermodynamic models that we present further on.

D. Nuclear Physics and Space Industry

The interaction of a charged particle beam with a material induces a dynamical equilibrium involving scattering, trapping, and radiation-induced conductivity (RIC). Surface potential is a tool to monitor the charging process as well as the charge decay after irradiation.

The pioneer in this domain was Gross [34]. From an early study of the effect of cosmic rays on insulators, he developed a new research area, where electron beams are used to investigate trapping and conduction properties of the materials.

Concerning potential decay experiments, we have to mention the works of Frederickson [1], [35], and in France, studies led at ONERA in Toulouse for CNES and ESA [3]–[5], [36], [37]. Models were produced by theoreticians, as Coelho in France [10], [11], Sessler in Germany [38], or Malecki in Poland [39] to account for potential decay after electron beam charging.

II. EXPERIMENTAL

A. Introduction

We shall consider mainly in the following the classical type of potential decay measurements, described Fig. 2. The surface is charged, and the potential decay is recorded following charging on the charged area, after a short transfer time.

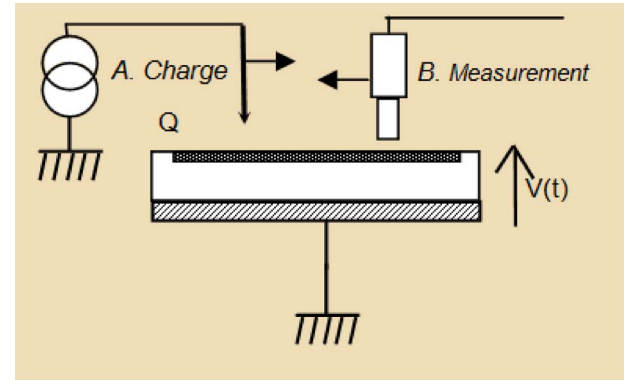


Fig. 2. Principle of potential decay measurements.

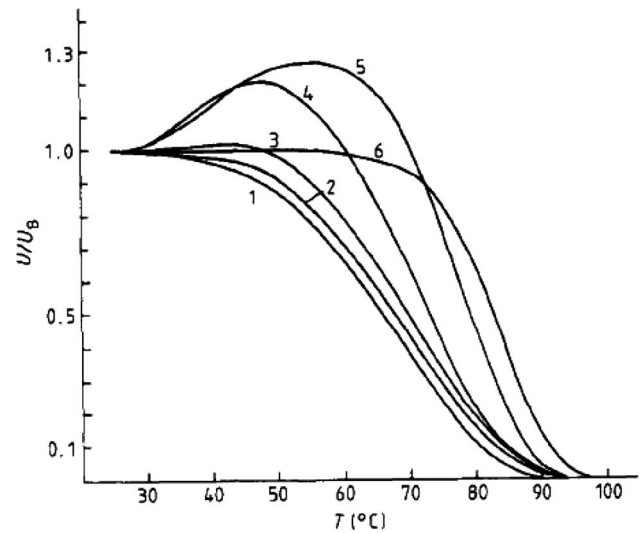


Fig. 3. Thermostimulated surface potential on polyacrylonitrile films showing an anomalous potential increase [42].

This experiment may be performed in isothermal conditions, or applying a thermal activation.

Thermostimulated surface potential (TSSP) measurements are closely related to thermostimulated currents [40]. The insulator is charged at high temperature, then cooled without removing the voltage, then gradually heated, the potential decay being measured instead of the current (Fig. 3).

TSSP has been used for polymer characterization since the early 1980s in Japan [8] and has also been used for studies on electret stability [41] or piezoelectricity [42]. Recently, it was proven useful for thin (less than 1 μ m) dye doped polymer film dielectric characterization [43].

B. Charging

The main charging processes of an insulator are by contact (either by a liquid or by a solid), corona discharge, and electron beam. From experiments, it has actually been known for a long time that potential decay may depend on the charging process, and not only on the amount of deposited charge, which reveals the role of the energy factor in charge injection phenomena.

Using an electron beam (typically from 1 keV to 50 keV), charge is usually directly injected into the bulk of the material, and the charge depth profile after beam cut off can be computed from various models. RIC and space charge effects have to be taken into account [44]. This injected charge will then drift into the insulator depending on the trapping and conduction properties of the material.

Contact deposited charge, on the opposite, is transferred on electronic surface states. Low-energy electron beam charging may as well lead to the deposit of a charge in surface states, either negative, or positive (secondary emission). The case of corona charging, charge being deposited as positive or negative ions, may appear very different. However, charge will also be transferred, at least partially, in electronic levels, through mechanisms that have been studied in detail on selenium by Vance [32].

The electronic surface states are usually deep traps in the forbidden energy band of the insulator, so that an additional energy will be needed to inject charge into the volume electronic states. When charge is performed by corona, this energy might be provided by the corona discharge itself. The deposit of a conductive paint on a polyethylene film, for instance, slows down the decay after corona charging [7]. The injecting effect of the corona discharge has been proven in 1977 [45] and later ascribed to the neutral particles of the negative corona discharge [20] and particularly activated nitrogen oxides [21].

Let us remember also that, even without energy activation, the injection of a trapped electron as a molecular ion cannot be ruled out. It is even a very likely phenomenon in thermoplastic polymers above a given temperature.

C. Measurements

Several kinds of probes might be used to measure the potential decay.

The simplest in its principle, the capacitive (or induction) probe is a field meter rather than a potential probe. It is made up by a conductive electrode on which an influence charge is induced by the studied surface. Influence depends on the probe distance to the surface; moreover, probe voltage remains in principle close to zero, so that it produces on the studied surface a field depending on the probe distance, which might perturb the potential decay process in a nonreproducible way.

This kind of probes (spheres and antennas) has been used, for instance, in the satellite SCATHA experiments [2]. This principle also allowed the development of a lateral high-resolution (about 80 1/4m) probe, which has been used by researchers in Bangor in the 1970s [45], [47], and led to good results for potential mapping. One can also mention the “field mill,” which allows, through the rotary motion of a star-shaped electrode behind a grounded electrode cut out with the same shape, to measure the absolute field, which avoids the inevitable drift of the capacitive probe [48].

The most common electrostatic probe model is however based on the vibrating reed (Kelvin) principle, with a feedback loop that allows setting the probe at the surface potential. The major advantage of the system is to perform voltage absolute measurement, with a minimum perturbation of the surface.

Decay models usually consider the electric field to be zero above the surface, which is effectively achieved using this kind of probe. The development of atomic force microscopy led also more recently to microscopic Kelvin probes that can be used to obtain high-resolution (up to 10 nm) maps of the surface potential distribution on nonconductive materials [49], [50]. In Kelvin probe force microscopy, the tip movement is analyzed by two feedback loops, the first to control the distance from the probe to the surface, and the second to minimize the electric field between them.

After charging the insulator by electron implantation in a scanning electron beam microscope, potential measurements can also be performed by measuring the deflection of a low-energy electron beam parallel to its surface [31]. The “electrostatic mirror method” can also be used for computing charge decay from the electrostatic mirror contraction, as described in [51].

III. GENERAL EQUATIONS

Let us consider a dielectric as a large plate of thickness d small compared to its other dimensions, so that side effects can be neglected. A 1-D model of the potential decay can then be proposed, where the charge densities, field, and potential are function of the distance to the ground only. The boundary conditions are that the lower surface is grounded, and it usually is assumed that the electric field is zero at the upper surface (which is indeed established in the Kelvin case). Considering, for instance, the electric displacement $D = \epsilon_0 E + P$ (P being the polarization inside the dielectric), its value in the dielectric close to the upper surface is then equal to the free charge density $q(t)$ deposited on this surface.

Modeling the potential decay experiment, the dielectric is usually assumed to be charged instantaneously at $t = 0$, and potential monitored immediately afterward. During the potential decay, we have an open circuit, so that the continuity equation can be written in any point of the dielectric, as

$$\frac{\partial D}{\partial t} + \left(\sigma + \sum_i \mu_i \rho_i \right) E = 0 \quad (1)$$

The first term is the time derivative of the electric displacement. The second term is related to the electric conduction, σ being the intrinsic conductivity, μ_i and ρ_i being the mobility and charge density of additional injected charge carrier, E being the electrical field. Diffusion is in this case assumed to be zero.

We will examine separately in next sections the influence of these three physical mechanisms on the decay.

IV. RELAXATION

Dielectric relaxation of most insulating materials, among them almost all the polymers, reveals a very low frequency component, linked as well to internal molecular reorganization (polymer chains slow movement) and to interfacial polarization phenomena. This response can be modeled in the linear regime by a dielectric response function $\phi_E(t)$, establishing a

convolution relationship between the electrical displacement and the field [52]

$$E(t) = \frac{1}{\varepsilon_0} \int_{-\infty}^t D(\tau) \phi_E(t - \tau) d\tau \quad (2)$$

In the case of a homogeneous dielectric, with no space charge, the electrical field is constant in the dielectric, and can be related to the potential by $E = V/d$.

Equation (2) can then be transformed to link measurable values

$$V(t) = \frac{d}{\varepsilon_0} \int_{-\infty}^t q(\tau) \phi_E(t - \tau) d\tau \quad (3)$$

Assuming then the conductivity to be zero, the continuity (1) implies that, during the decay, $\partial D / \partial t = 0$, so that the free charge density on the surface is a constant. It can be written $q(t) = \Gamma_0(t) q_0$, $\Gamma_0(t)$ being the Heavyside function centered on $t = 0$.

From (3), we can deduce

$$\frac{dV(t)}{dt} = \frac{dq_0}{\varepsilon_0} \frac{d}{dt} \int_0^t \phi_E(\theta) d\theta = \frac{dq_0}{\varepsilon_0} \phi_E(t) \quad (4)$$

This relationship implies that the material does not react instantaneously to the changes of the electric field. If a charge is deposited on the surface at a given time, a potential step will follow, linked to Poisson relationship, screened by instantaneous polarization processes, and then, potential will decrease under the influence of slow polarization. The general theory of this phenomenon was established by Gross in 1942 [27], and a recent model has been given in [53].

For reasons studied in particular by Jonscher [54], the usual response function in condensed matter is composed of time power laws. In this case, potential derivative, proportional to $\phi_E(t)$, will also be composed of time power laws, with different coefficients [12], [53], [55], which will appear as line segments or asymptotes on $\log(dV/dt) = f(\log(t))$ plot, with slopes more or less close to unity. We have been publishing examples of this kind of behavior in epoxy [12] (Fig. 4), polypropylene [13], [14], and polyethylene [13]. Das Gupta [47] also attributed decay on polyethylene films to slow polarization phenomena.

These relaxation phenomena, either dipolar or interfacial, might be negligible in many highly stable materials used in the space industry, as fluorinated polymers or polyimide (PI). However, they cannot be completely ruled out, and their existence has to be checked carefully.

V. INJECTION OF THE DEPOSITED CHARGE

A typical experimental evidence of charge injection into the bulk is given Fig. 8 on Teflon FEP by charge mapping measurements [38] (see Section V-E). Many models are based on this injection hypothesis and on various physical hypotheses concerning transport. The occurrence of crossover (Fig. 1) is a direct proof for a nonlinear behavior of the dielectric and

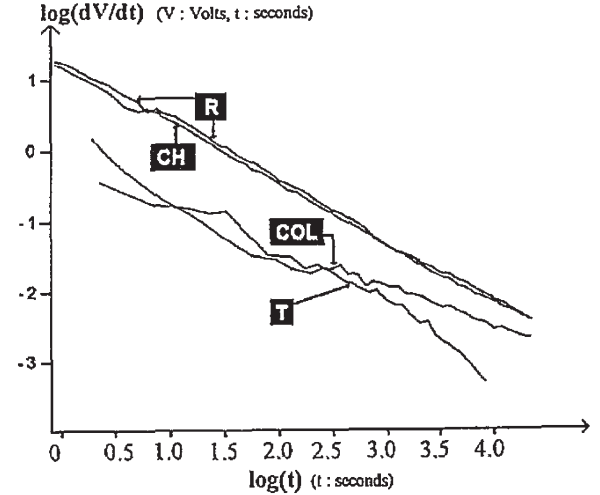


Fig. 4. Time derivative of the potential in a log-log plot, on epoxy resin charged at low fields (200 V.mm^{-1}) [12]. Labels are referring to various compositions: T: bisphenolic epoxy resin, basic, COL: red colored, CH: with SiO_2 filler, R: red colored and with SiO_2 filler.

can be considered as a serious indication for charge injection effects [15].

A. Electrostatic Models

We classify under this designation transport models which basically do not take into account the influence of charge trapping and detrapping processes on the charge decay mechanism, considering therefore transport only by means of a carrier mobility μ . Within this theoretical frame, from the boundary conditions, various models can be established. Detailed calculation has been established by Batra *et al.* [30], who computed in the general case decay equations. The main idea, in the “Kelvin” case of a zero field outside the dielectric, is that charge distribution may be considered as successive sheets, parallel to the surface, driven by a constant field along their drift through the material, so that each of them moves forward at constant speed (even if mobility is field dependent). If charge injection from surface to bulk electronic states is assumed to occur during the charging phase, these models therefore predict a linear potential decrease as long as the charge front has not reached the grounded counterelectrode; they also imply that the initial potential decay rate is proportional to the square of the initial voltage. After the charge front transit time, the voltage versus time plot presents a bend, the number of charged sheets contributing to the decay decreasing gradually. The last charged sheet is driven by a zero field and hence moves at zero speed, so that it will stay on the upper surface.

This kind of model was also developed by Wintle [17], who included diffusion and conduction, and later by Coelho [9], [11], Wintle model being then developed by Hill [56]. However, despite the interest of these models for semiconductors, their application field seems to be quite limited concerning insulating materials, since we were not able to find any paper reporting in an indisputable way, on real dielectrics, a behavior corresponding to these calculations. What is usually found is shown on the upper part of Fig. 6, so that the linear initial decrease that the

models predict, which would lead to a horizontal straight line on the plot, is never observed. Let us remark that taking into account a field depending mobility [33] does not change this prediction of the model.

Assuming a steady-state equilibrium to be achieved, and considering the mean time τ_t spent by the charge in trapping levels, compared to the duration τ_0 of its stay in transport levels, it is also possible to define an effective mobility taking into account trapping processes, which will be $\mu_{eff} = \mu\tau_0/(\tau_t + \tau_0)$ [57]. Nevertheless, it obviously does not change anything to the shape of the $V = f(t)$ curve, produced by the model.

B. Volume Trapping and Dispersive Transport

A correct treatment of the potential decay problem in a disordered material actually requires that trapping during charge drift should be taken into account, since it will greatly influence the potential decay shape. A first line of reasoning, modeled by Wintle [18] (and taken up later by [29]), considers an effective mobility for carriers, though restricting it to hopping through transport levels between shallow traps only, while irreversible trapping in deep traps is introduced. This kind of models predicts a decay tending toward a constant, nonzero value, of the potential, and is closer to experimental results in dielectrics than the models reviewed above.

Later on, Von Berlepsch [23] produced a more complete treatment for this problem by introducing reemission of charge by deep trapping levels, together with field-assisted surface charge emission according to Sonnonstine's model. He also defined the concept of a time-dependent effective mobility. Recently, experimental data on polypropylene were confronted with a model and computations along the same lines [58].

However, it is not possible nowadays to treat the question of charge transport in a disordered medium, without mentioning the more general frame of dispersive transport, which handles the question as a whole, by assuming a continuous energy distribution of the trapping levels. Since Scher and Montroll paper [59], many transport models in disordered systems have been proposed. It is possible to consider numerical computing of this multiple trapping problem (an example can be found in [31]).

A simplified approach introduces the concept of demarcation energy, defined as

$$E_d(t) = kT \ln [(N_c/N_t)(t/\tau_0)] \quad (5)$$

(N_c being the conduction states density, N_t the trap density, and τ_0 the carrier lifetime in conduction states).

The hypothesis is that, at a particular time t , the emission of traps whose energy is below E_d can be neglected, whereas one can consider that higher energy states are in equilibrium with transport states. An analytical computation of the potential decay along these lines was performed by Arkhipov [60] (see also [61]). Assuming an exponential shape of the trap distribution, he showed that a $\log(dV/dt) = f(\log(t))$ plot was tending toward two straight lines with different slopes, on either sides of a mean transit time value (Fig. 5).

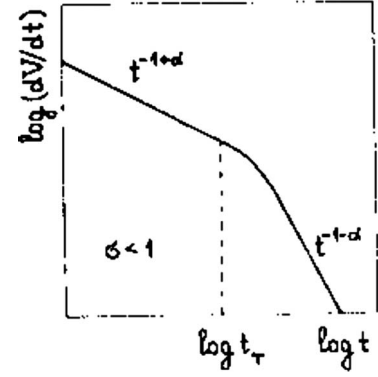


Fig. 5. Potential decay due to dispersive transport [59].

However, one has to remember that this approximation, assuming equilibrium between shallow states and transport states, relies on the fact that the lifetime of carriers in transport states is negligible compared to the dwell time at the demarcation energy, so that the model is not valid at short times, when this equilibrium is not reached.

C. Detrapping Controlled Decay

If the carriers transit time is low compared to their mean characteristic detrapping time, the geometrical factor can be completely eliminated from the models, detrapping kinetics being then the main factor determining the shape of the decay, rather than charge displacement. Trapping charge stability is proportional to the exponential of the trap energy depth, so that the slowdown observed in the potential decay rate can be attributed to the progressive increase in the trapping energy depth of the carriers. The demarcation energy considered above can thus be considered as the limit between filled deep traps and shallow traps, emptied since the beginning of the potential decay. Using this approximation, formulated by Simmons and Tam [62], and applied by Watson [31] to potential decay modeling, detrapping emission current is linked to the trap energy density $N(E_d)$ at the level $E_d(t)$ in the following way:

$$I(t) \propto N(E_d(t)) \frac{dE_d}{dt} = N(E_d(t)) \frac{kT}{t} \quad (6)$$

One can deduce from this equation that the plot of time multiplied by intensity $tI(t)$, versus the time t , will give using a semilogarithmic scale an image of $N(E)$, the trap density versus their energy depth. Concerning the potential decay experiment, the decay rate dV/dt will be proportional to the right part of (6), so that, multiplied by time (tdV/dt), it will give the same image of $N(E)$. Watson has shown that this simple plot, compared to a detailed numerical calculation, underestimated the shallow traps density compared to deep traps, since it neglects the influence of retrapping during the charge drift. However, it can give interesting results, which he presented on monodisperse polystyrene. On Fig. 6, we reproduced from [31] the potential decay rate recorded on this material, and a $tdV/dt = f(\log(t))$ plot, whose abscissa has been converted in energy using (5), assuming the best value of $(N_c/N_t)(t/\tau_0)$ to fit the curves at various temperatures.

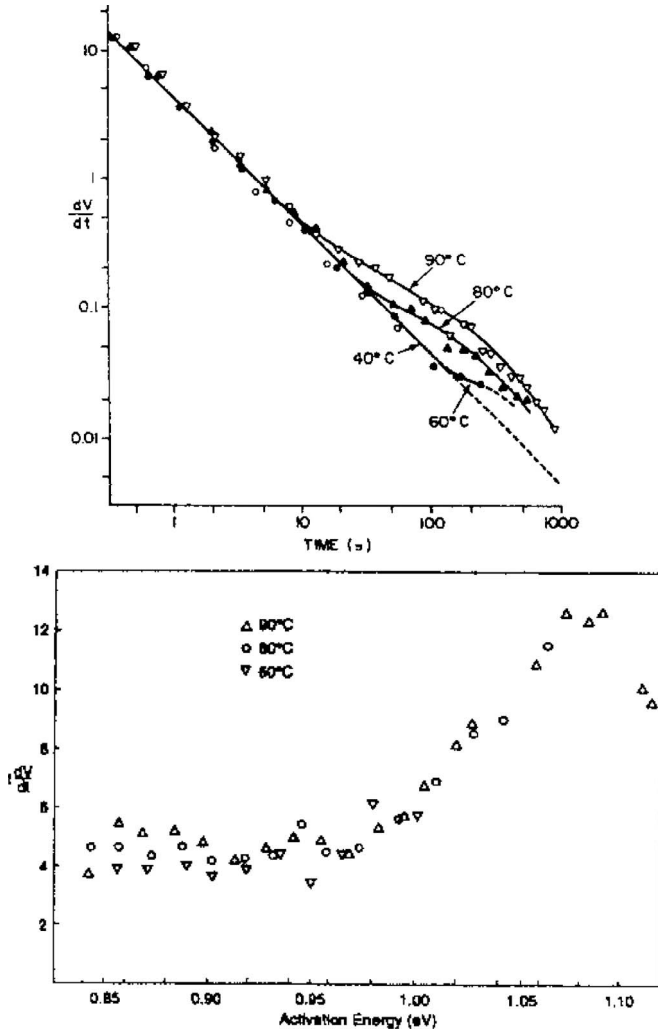


Fig. 6. Data treatment according to the demarcation energy model [31].

D. Surface Trapping

As we mentioned above (Section II-B), deposited charge is quite often trapped in surface energetically deep levels. A coherent explanation of the cross-over phenomenon has been given by Sonnonstine and Perlman [19], assuming partial surface charge retention, surface emission into the bulk being determined by an injection law depending on the field. According to their calculations, two successive regimes occur: a high-field regime, limited by the drift of the internal space charge, followed by a low-field regime, limited by the surface charge retention, the crossover being due to the existence of the first regime only. The classical dielectric model assumed by this computation is given in Fig. 7. This kind of modeling introduces in potential decay models the notion of surface energy barrier and field-assisted injection over it. We already mentioned this point in Section II-B.

This is in good agreement with thermostimulated current measurements which, according to ideas initially expressed by Von Seggern [63], may allow discriminating surface from volume traps and have shown that, on most of insulating materials, surface traps were deeper in energy than volume traps. Therefore, situations where surface progressive detrapping

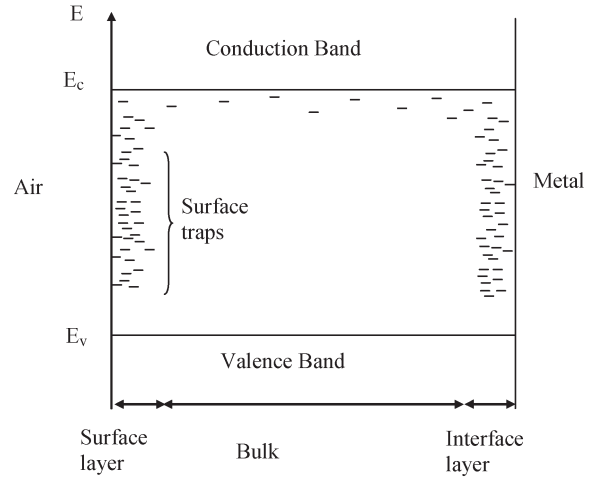


Fig. 7. Classical model for charge trap energy and geometric distribution in the insulator.

ping is determining the dynamics of the decay on the long time should be numerous, though this hypothesis is practically never suggested in the literature. In a recently published article ([14], see also [13]), we have shown that a model of progressive surface detrapping was the only way to account for measurements performed in Supec on 6 μm polypropylene films, negatively charged by corona, at high fields and ambient temperature.

These models were mainly developed to account for corona-charged experiments, but they might also have a validity for electron beam-charged polymers, either when the beam energy is low, or when secondary surface charge is injecting into the bulk and contributes to the voltage decay. In addition, Von Seggern experiments suggest that the depth of the deep trapping surface region can extend to 5 μm [63].

These surface injection models may of course also be applied to charge injection from the counterelectrode. The interface and energy barriers are then different, and polarity of the carriers opposite, but the same kind of models may be applied.

E. Application to Fluorinated Ethylene Propylene (FEP)

An experimental illustration of charge injection can be given on electron beam charged Teflon FEP. Using LIPP (laser-induced pressure pulse) or other charge mapping techniques, it is possible to determine the mean profile of the charge density inside the dielectric. The evolution of this profile (Fig. 8) with annealing showed that surface potential decay at 120 $^{\circ}\text{C}$ could be ascribed to the injection of the negative charge [38]. The conduction mechanism involved according to the authors is detrapping accompanied by fast retrapping of the charges.

At ambient temperature, however, on this material, it is usually assumed that electrons are deeply trapped, while holes are the only mobile carriers [64], [65]. Their mobility is limited by trapping, with trapping levels between 0.7 to 1.5 eV [66].

When Teflon FEP is positively charged by corona, the surface slowly discharges, but part of the charge is retrapped by volume deep trapping. This trapped charge seems to be more stable than the surface positive charge, which leads to a different picture

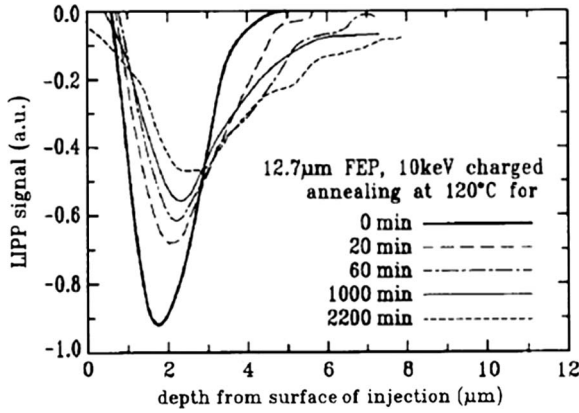


Fig. 8. Change of the charge distribution in FEP after annealing. Sample charged with 10-keV electron beam through left front side [38].

than Fig. 7. Annealing at 140 °C may detrapp this volume charge. Charging by a negative corona leads to a very stable surface potential, except at high fields and temperatures [65].

VI. MODELS INCLUDING OTHER SOURCES OF MOBILE CHARGES

A. Homogeneous Conduction Models

Assuming that polarization is stabilized, so that the material can be modeled by its constant permittivity ϵ , and neglecting space charge effects, the existence of an intrinsic conductivity σ transforms (1) into $\sigma E + \epsilon(\partial E/\partial t) = 0$, so that

$$\frac{dV}{Vdt} = -\frac{\sigma}{\epsilon} \quad (7)$$

A constant intrinsic conductivity will then lead to an exponential shape of the potential decay, with a time constant σ/ϵ . The conductivity being proportional to the product of the mobile charge carrier density by their mobility, potential decay models assuming intrinsic conductivity rely on particular hypotheses on carrier generation, recombination, and mobility.

We will underline here two important facts: first, the existence of a nonzero conductivity in a material will lead any charge present in the material to be screened by conduction carriers with the σ/ϵ time constant, which is hence the time value above which the material can be assumed to be conducting. Fig. 11, in our opinion, shows this effect on PI (see Section VI-D).

Second, due to their very low volume conductivity, surface conduction often predominates on polymers. Charge lateral spreading is frequently observed [46]. Thus, several potential decay models on dielectrics take into account this parameter. They were developed for surface-aged insulators in the presence of humidity [67]. However, in this case, a truly conductive surface layer is involved, which properties are very different from the material itself. Another situation, found when a slightly conductive surface layer is produced by low-energy irradiation, will be described Section VI-C3.

B. Carrier Generation and Recombination

Charge carrier generation can be due to other processes than injection from the surface or the electrodes. Electron-hole pairs can be created through band ionization, which, in insulating materials with gaps greater than 5 eV, can only be performed by irradiation. Charge detrapping from shallow traps can also be assisted by temperature and field through Poole-Frenkel effect at high fields. That will lead to a field-dependent conductivity, and potential decay has been modeled according to this mechanism [68]. An analog process is ion pair creation through field or temperature-assisted dissociation of impurities.

Carrier recombination may occur in volume, by electron-hole recombination or ion pair formation. For instance, in Teflon FEP, trapped electrons are acting as recombination centers for mobile holes. Another recombination mechanism occurs at the electrodes. According to Malecki [69], the long-term stability of Teflon FEP electrets is due to the exhaustion of the intrinsic carriers assisted by the field. Then, the charge or potential decay rate will be determined by the carrier creation rate only. It will hence increase linearly with the thickness of the sample and will tend toward a constant for long time periods (months or years).

C. Radiation-Induced Conductivity

Introduction: RIC in insulating materials has been widely studied since pioneering works of Gross [34], [70] and Fowler [39], [70]. We will here limit the application of the RIC models to the modeling of potential decay experiment. In most cases, the RIC is not constant in time, and its effects are not homogeneous in space. In a classical experiment under a constant voltage polarization (current measurements), RIC may for instance increase in a first phase and then reach a maximum value, followed by a slow decrease [70]. After radiation stops, a delayed RIC (DRIC) persists [34], [70]. The RIC decay can be accelerated by the field [39] which drains the charge carriers.

The dynamic character of the potential decay experiment, due to the charge evolution in time, will be complicated by this dynamic evolution of the RIC. Simple models may, however, be built in some cases.

Homogeneous and Stable RIC: RIC is homogeneous when the absorption of the radiation by the dielectric can be neglected. A homogeneous and stabilized RIC leads to a constant and homogeneous conductivity within the material, depending on the dose rate

$$\sigma_{RIC} = K \left(\frac{dD}{dt} \right)^\Delta \quad (8)$$

K and Δ being material-dependent parameters and D the dose [69].

The potential decay may then be very easy to model and a powerful tool to measure this conductivity. The conductivity may then be deduced directly from the potential decay rate by (7), as has been done in Fig. 10 [4], [5].

Surface RIC: The opposite situation, assuming a short distance penetration of radiation into the dielectric, can lead to a lateral spreading of the charge if the dose is high enough to generate surface conductivity [37]. Radiation may also act as

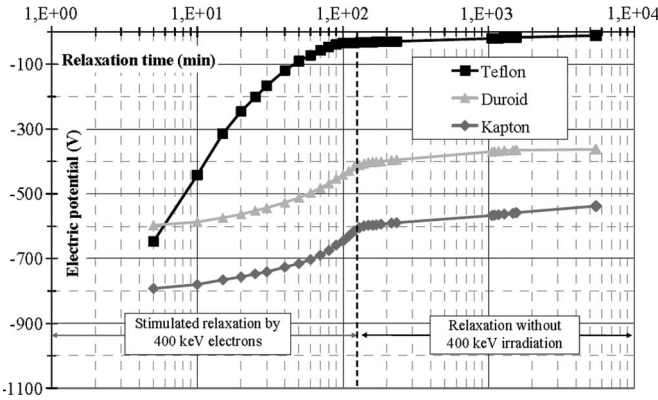


Fig. 9. Potential relaxation on several materials in the ONERA SIRENE facility after initial 20-keV electron charging and stimulation [5].

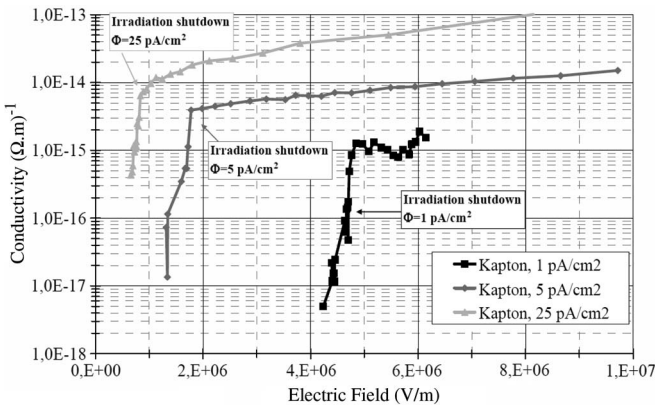


Fig. 10. Calculation of conductivity from SIRENE experiment for Kapton subjected to three intensities corresponding to 2.4, 12, and 60 Gy.s^{-1} dose rates [5].

a charge injecting factor when charge is deposited in surface levels (by corona, for instance). A model of the decay following photoexcitation of the carriers has for instance been developed by Batra for xerographic applications [30]. This model assumes that light generates electron-hole pairs in the bulk close to the corona-charged surface. They will be polarized by the induced field, the corona charge being thus neutralized by the carriers (hole or electrons) of the opposite polarity, while the carriers of the same polarity will drift into the material.

D. Application to FEP and Polyimide

Potential decay experiments under irradiation by a penetrating 400-keV electron beam have been performed on charged Teflon FEP and Kapton PI, as can be seen for instance in Fig. 9 [5]. The ionization effect of such a beam is more important than its charging effect. The influence of the radiation-induced conductivity on the potential decay is here clear, and a constant RIC model may apply. From the voltage decay, conductivity has been computed for different dose rates (Fig. 10).

The effect of the field is, however, more difficult to deduce from these curves, since each of them is built from one potential decay experiment only, so that time increases while the field diminishes, and trapping effects might interfere. After beam cutoff, the decay is much slower, though DRIC is not negligible.

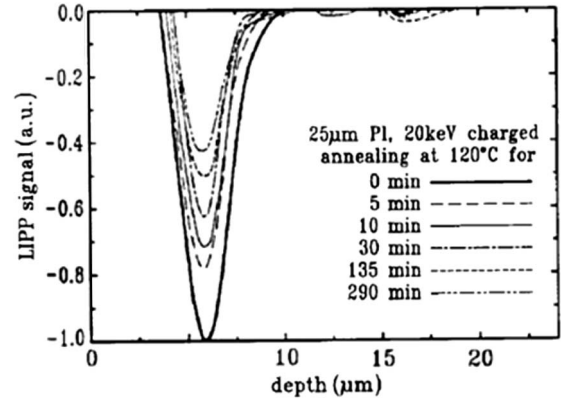


Fig. 11. Change of the charge distribution in polyimide after annealing. Sample charged with 20-keV electron beam through left front side [38].

A striking feature of these materials commonly used as thermal blankets or coatings is the diversity of their ageing behaviors, which leads after a given irradiation time to opposite results to what is presented Fig. 9. Kapton do not charge anymore after a long period in flight, due to the elevated induced conductivity initiated by space radiation, while FEP can reach high charging potentials due to an elevation of its electric resistivity under high radiation dose [5].

The influence of gamma ray-induced conductivity on potential decay on PI was also studied in detail by Frederickson [35].

On PI, mechanisms of mobile charge generation exist even without any ionizing radiation exposure. Fig. 11 shows LIPP measurements after 120 °C annealing on electron beam-charged PI: a very different evolution of the space charge is observed compared to FEP [38]. Our analysis of this result is that the charge is progressively screened by an intrinsic conductivity, maybe due to photoconduction (though the authors do not mention this possibility in their paper).

VII. SURFACE NEUTRALIZATION OR EMISSION

In some particular contexts, on highly insulating materials, the potential decay may be mainly due to a current flowing in the space surrounding the dielectric. Ion drift in ambient air, usually negligible at low fields, might play a critical role in particular environments, as large HVdc apparatus, as has been reported by Kindersberger [71]. Coelho also favored this explanation to account for surface potential decay experiments on FEP [10].

Another mechanism has been imagined to account for the rapid initial potential decrease on Teflon FEP charged in vacuum by an electron beam [4], [36]: the electronic emission continuing after beam cutoff, due to an inversion layer at the dielectric surface (Malter effect).

VIII. CONCLUSION

A great variety of mechanisms has been presented. Choosing a model is difficult since three different classes of physical phenomena can produce the same power-law time dependence for the decay rate, leading to straight lines on the $\log(dV/dt) = f(\log(t))$ plot, with a mean slope close to -1 , and a bend making possible to distinguish two domains, with different

slopes. This bend could be interpreted as the center of a dipolar relaxation time distribution, in the frame of a model where deposited charge does not inject (Fig. 4), or as a transit time in a transport model (Fig. 5), or as a peak of a trap energy density distribution in a thermodynamical model (Fig. 6).

Furthermore, several phenomena might superimpose. The global material response is generally depending on multiple factors. The material displays a specific order at the atomic scale (more or less deducible from its chemical formula), a structure at the nanometric scale due to the interactions between neighboring molecules, defects and impurities, a structure at the micrometric scale (spherulites for polyethylene, grains for polycrystals), together with surface phenomena, free surfaces, voids, and possibly interfaces when dealing with composite materials. Phenomena at each of these scales will have an influence on the general behavior of the dielectric, at least in given conditions of field, pressure, or temperature.

Practically speaking, we suggested elsewhere [13], [14] to systematically choose representations giving prominence to the decay dynamics, and we proposed the use of the tdV/dt versus $\log(t)$ plot, which has been developed within the frame of demarcation energy models, as being the most appropriate for the polymers used in insulation or in electrets, independent of any hypothesis on the physical mechanism involved (for instance a bend or a peak can be a transit time, or a particular characteristic relaxation time). Using $V = f(t)$ plot, it is much too easy to perform a very good fit of a model curve with the experimental one; hence, that kind of representation is not satisfactory.

Multiple experiments have to be performed, involving changes of the interfaces, of the thickness, changing the voltage and polarity to check what kind of model (polarization, transport, detrapping, or surface conduction) is appropriate. Potential scans can help to evaluate the importance of charge lateral spreading, or surface conductivity, and return voltage measurements can also provide a lot of information [12], [16]. Associating charge mapping techniques giving a depth profile of the trapped charge, as LIPP, or thermal step method, or pulsed electroacoustic, etc., is also necessary to confirm the physical hypothesis expressed in the models [16], [36], [38], [72].

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